

Introduction

- Deep reinforcement learning agents commonly use high-dimensional neural representations, even when solving tasks whose underlying decision structure is comparatively simple.
- The manifold hypothesis suggests that task-relevant structure may lie in a much lower-dimensional subspace.
- Such structures are usually discovered through optimization, encouraged by auxiliary objectives, or recovered post hoc.

Research Question

Can we impose such a low-dimensional structure and learn directly within it?

Orthogonal Bottlenecks

- An encoder produces features $z = \phi_\theta(s) \in \mathbb{R}^D$.
- A fixed orthonormal matrix $B \in \mathbb{R}^{D \times k}$ with $B^\top B = I_k$ projects these features into a low-dimensional representation $h = B^\top z \in \mathbb{R}^k$.
- The policy and value heads receive only h , the underlying RL algorithm and training objective remain unchanged.

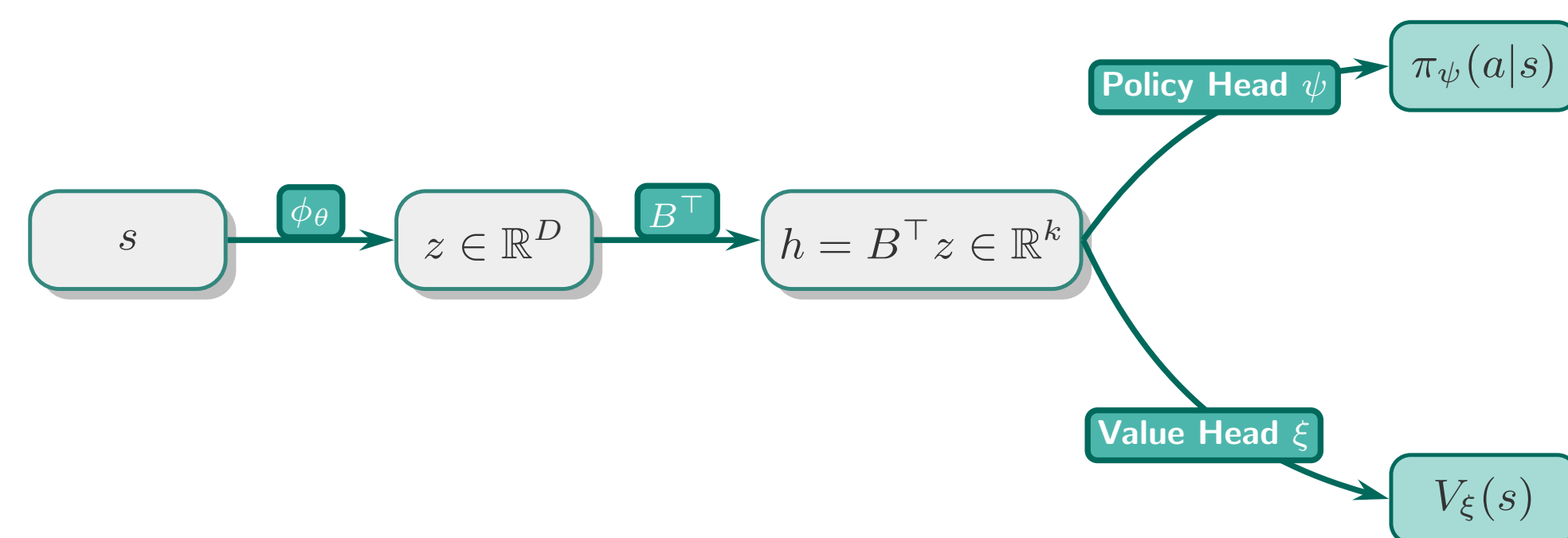


Figure 1: The orthogonal bottleneck adapted to an Actor-Critic architecture.

When is the Bottleneck Sufficient?

- Suppose the optimal value function is linear in the learned feature space with $V^*(s) = \Theta^* \phi(s)$.
- Let $r = \text{rank}(\Theta^*)$ be the intrinsic dimension of the value function in feature space.
- If $k \geq r$, then a fixed orthogonal bottleneck can still exactly represent and learn V^* .

Theoretical Message

If the bottleneck dimension is at least as large as the intrinsic rank of the task, no loss of expressivity occurs and learning is still possible.

Low-Dimensional Value Manifolds

With representations often compressed to single-digit dimensions, in some environments, we can directly visualize the learned value manifolds.

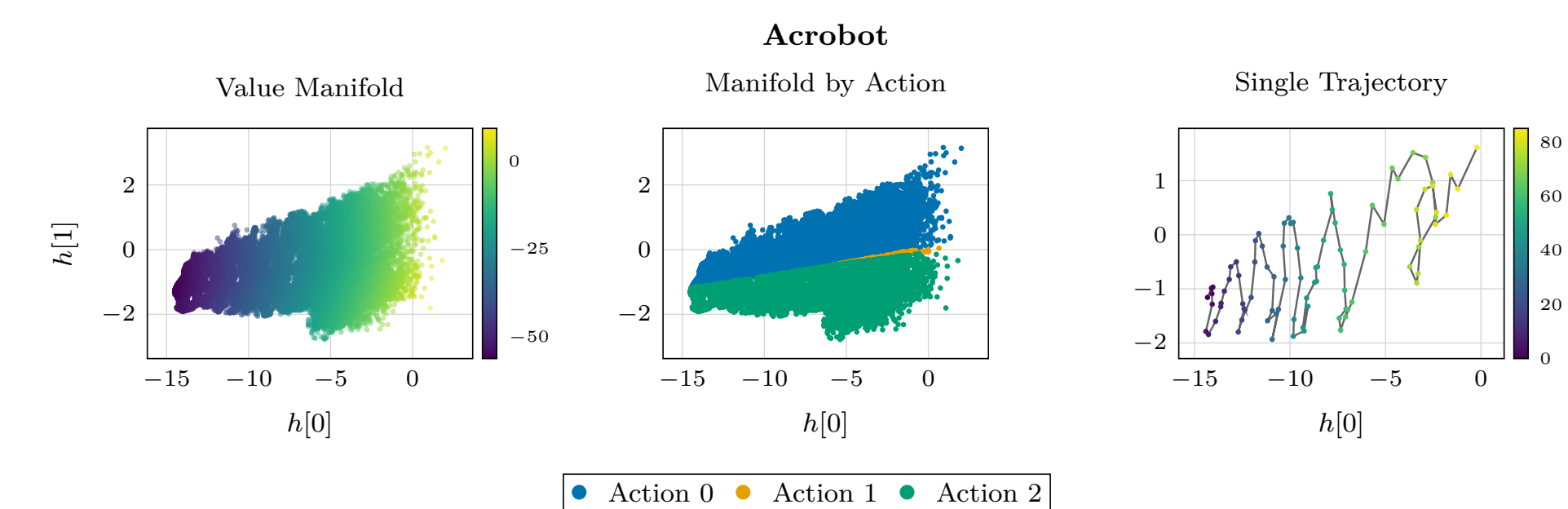


Figure 2: Two-dimensional bottleneck representations. Points correspond to visited state-action pairs, colored by value, action, or timestep.

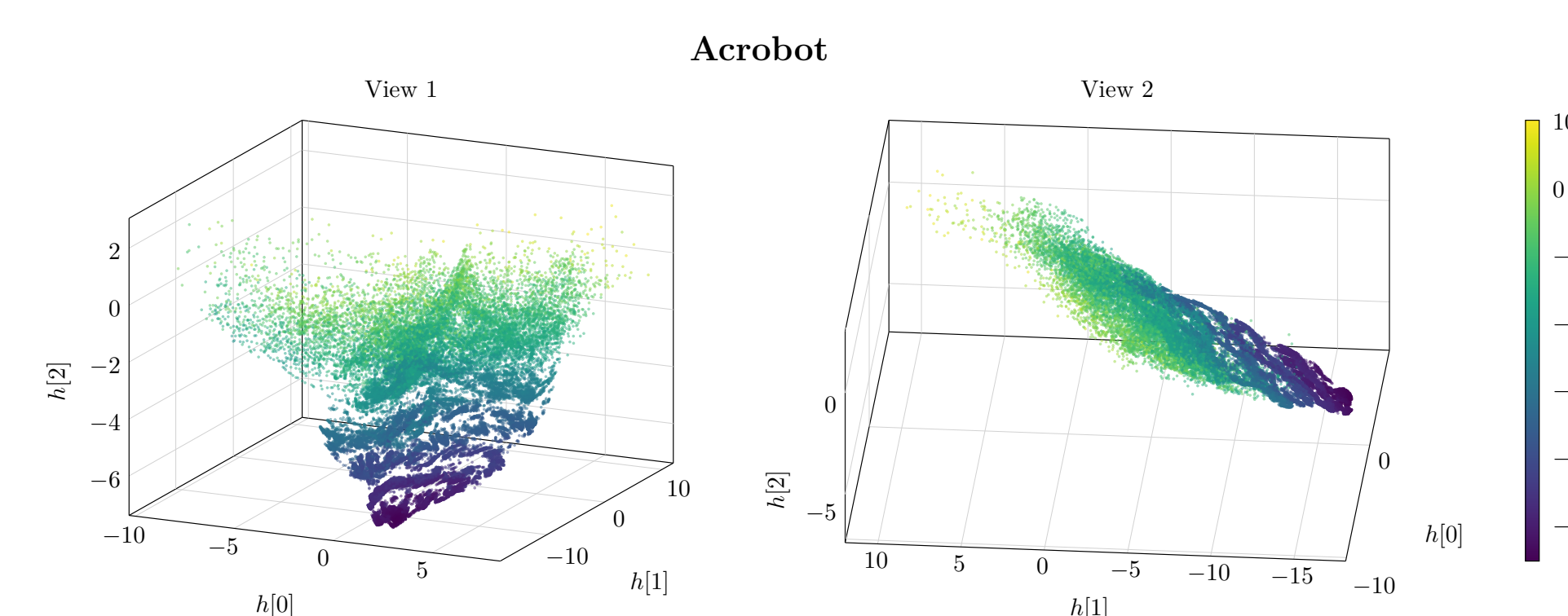


Figure 3: Three-dimensional bottleneck representations. The embeddings remain nearly flat rather than filling the full ambient space.

Performance Recovers at Small k

Across larger benchmarks, performance degrades when the bottleneck is too small, but recovers once k exceeds a small task-dependent threshold.

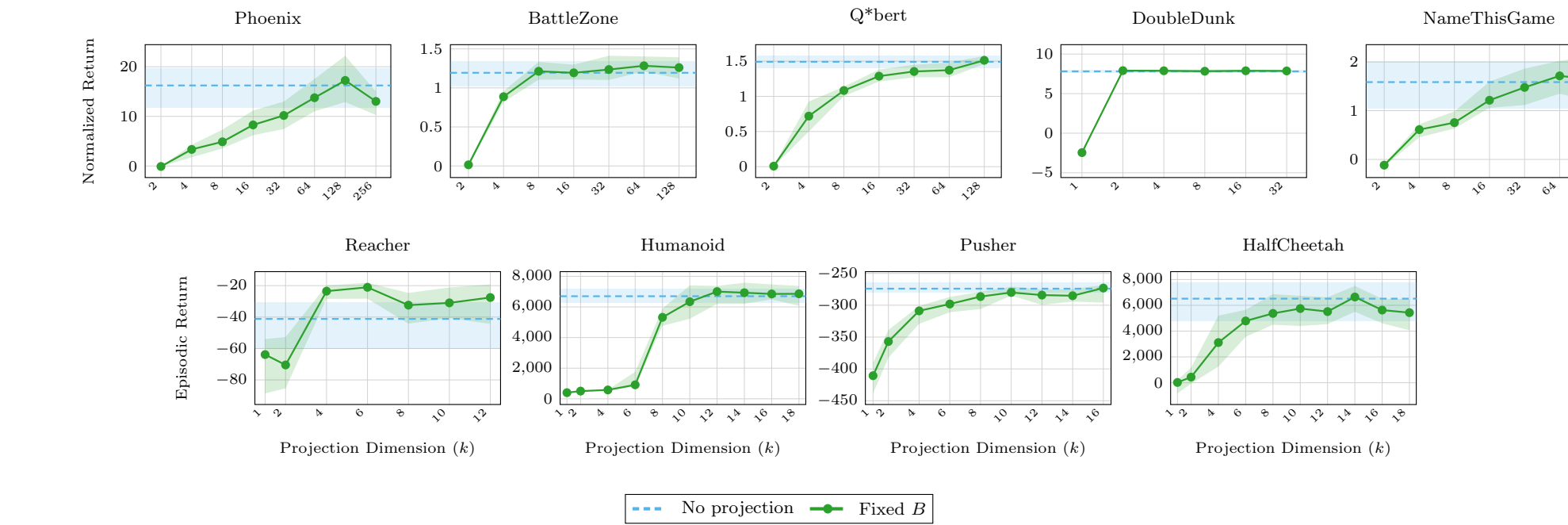


Figure 4: Final performance as a function of bottleneck dimension k . After a small recovery threshold, increasing k gives little additional benefit.

Orthogonality, To Train or not to Train?

What if we learn the bottleneck instead of fixing it?

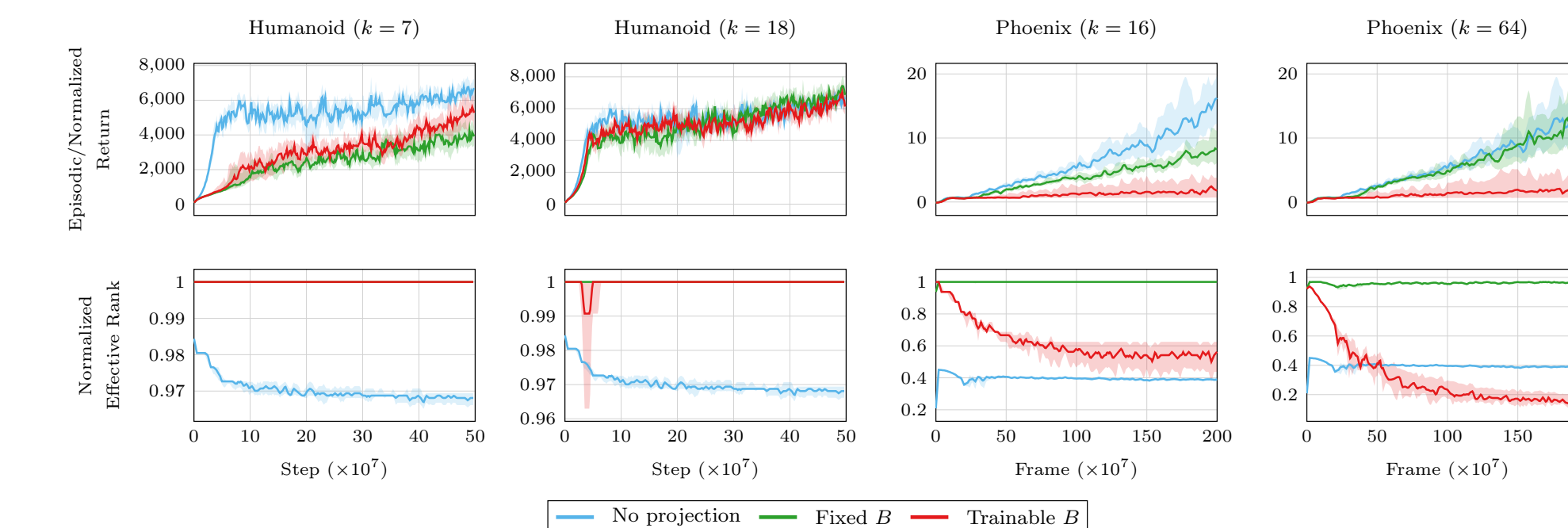


Figure 5: Trainable projections can be beneficial or unstable depending on the environment, with performance degradation coinciding with severe effective rank collapse.

What if the bottleneck is fixed, but not orthogonal?

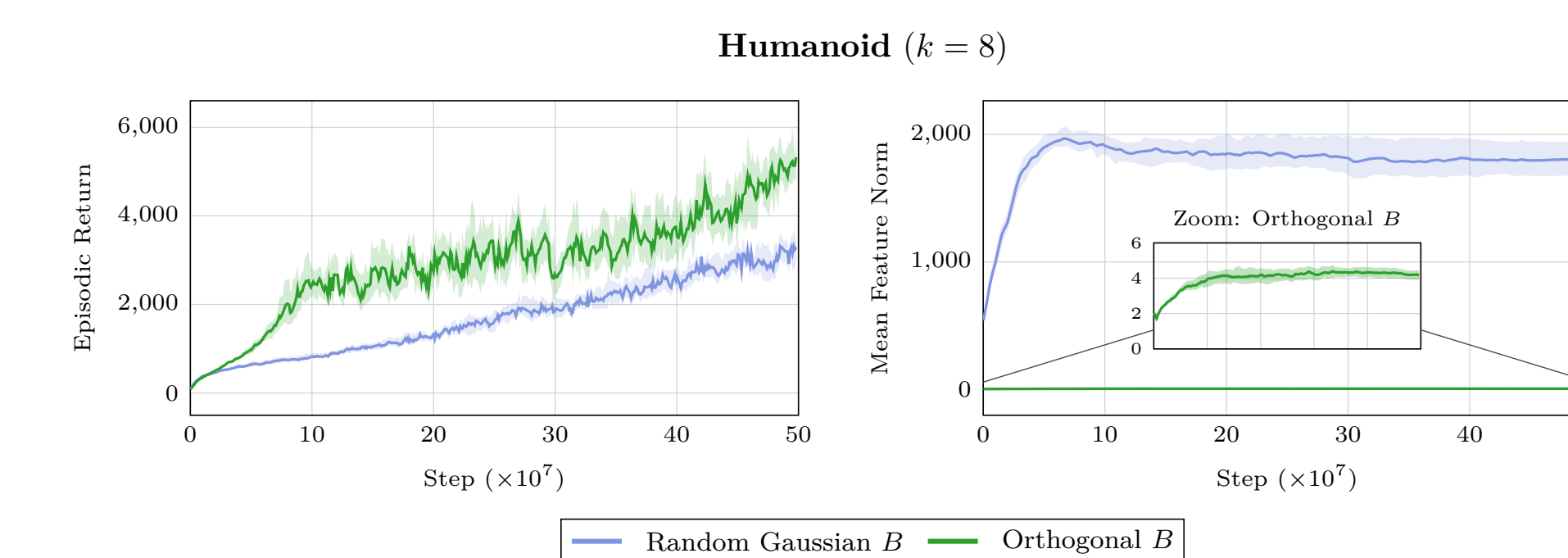


Figure 6: Feature norms explode when using a fixed Gaussian projection and lower performance is achieved, while a fixed orthonormal B learns reliably.

Shared Bottleneck for Multi-task RL

A single low-dimensional orthogonal subspace can also be shared across multiple tasks.

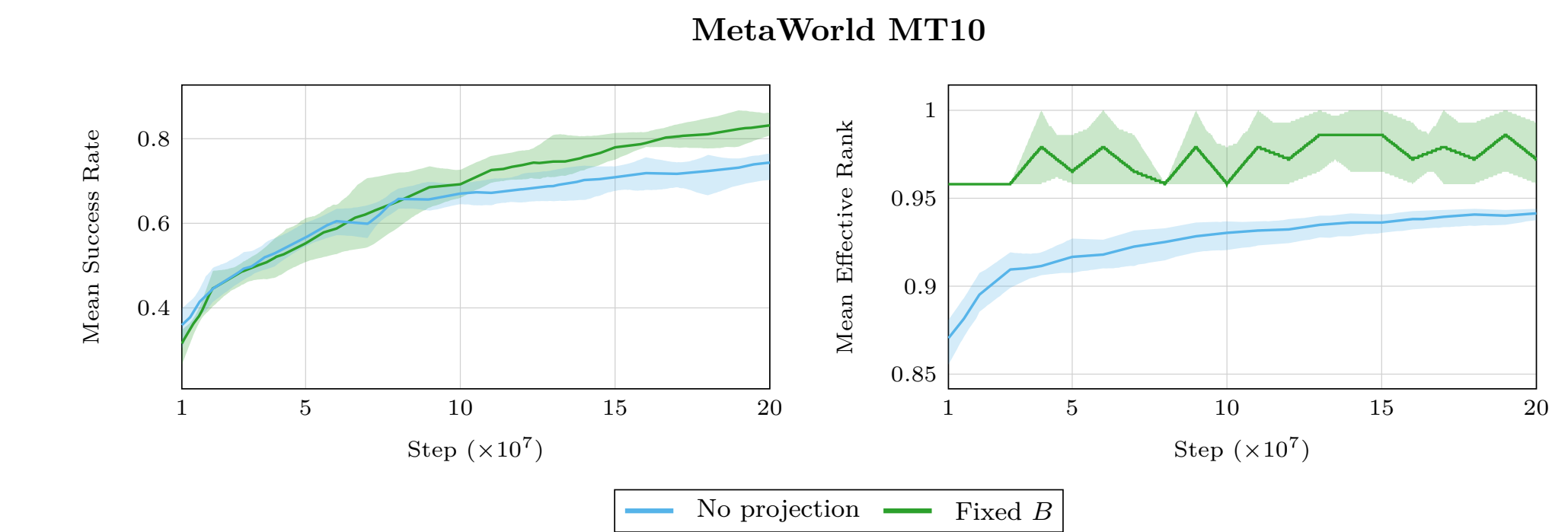


Figure 7: The fixed orthogonal bottleneck modestly improves performance while maintaining higher normalized effective rank.

Discussion and Conclusion

Main Message

Deep RL agents can learn directly within very low-dimensional subspaces, achieving extreme representation compression without loss of performance.

- Our results support the conjecture of Tenedini et al. (2026) that intrinsic manifold dimensionality is driven primarily by environment complexity rather than network size, though for value representations rather than policy manifolds.
- The theory assumes a fixed feature space where the value function is linearly realizable, while the experiments learn this space end-to-end.
- Still, the consistent recovery of performance after a small task-dependent threshold in k suggests that learned deep RL representations may contain approximate low-rank linear structure, making them more amenable to such analysis than is commonly assumed.

References

Tenedini, D., Zamboni, R., Mutti, M., and Restelli, M. (2026). From parameters to behaviors: Unsupervised compression of the policy space. In *The Fourteenth International Conference on Learning Representations*.